

Survey of Long-Sequence Time-Series Forecasting by Deep Learning Algorithms

Abstract—Long-sequence time-series forecasting (LSTF) has become a central problem in modern data-driven decision systems, as many real-world applications require exploiting extended historical contexts and producing reliable long-horizon forecasts under non-stationary, noisy, and often incomplete observations. Recent deep learning research has responded with a diverse set of algorithmic strategies, ranging from architectural reconfigurations of Transformers and attention, to frequency- and multi-resolution decomposition, graph-based dependency learning for large-scale multivariate systems, and new learning paradigms that leverage foundation models or synthetic priors for improved transfer and few-shot generalization. This survey reviews representative methods along these routes and distills several recurring patterns: (i) “where” and “what” a model attends to (time tokens vs. variate tokens vs. discrete latents) is increasingly treated as a first-order design choice; (ii) multi-resolution structure—implemented through wavelets, spectral operators, or progressive generation—has emerged as a unifying mechanism for stabilizing long-horizon extrapolation; (iii) multivariate dependency modeling is shifting from static correlations toward time-varying, scalable, and data-quality-aware structure learning; and (iv) robustness and uncertainty estimation are moving from auxiliary add-ons to core objectives for deployable forecasting. By organizing the literature around these methodological trajectories, this survey aims to clarify how recent systems trade computational efficiency, inductive bias, robustness, and transferability when forecasting long sequences.

Index Terms—Long-sequence time-series forecasting, long-term forecasting, deep learning, Transformers, attention, multi-resolution modeling, frequency-domain analysis, graph neural networks, foundation models, robustness, uncertainty quantification.

I. INTRODUCTION

Long-sequence forecasting is difficult not merely because the input window is long or the prediction horizon is extended, but because the statistical and computational pathologies of time series amplify with sequence length. Long contexts increase the burden of capturing long-range dependencies while avoiding spurious correlations, and long horizons exacerbate error accumulation, distribution shift, and sensitivity to rare events. Practical deployments further complicate the setting: multivariate signals may involve heterogeneous physical meanings across channels, missing observations arise from acquisition constraints, and anomalies or extreme values can dominate downstream risk. These realities motivate algorithms that remain accurate at scale, stable under non-stationarity, and reliable in terms of uncertainty.

Deep learning for LSTF has evolved into several identifiable technical routes. Transformer-based forecasters have been widely adopted due to their capacity for modeling global dependencies, yet quadratic attention complexity and representation choices can hinder long lookback windows and high-dimensional multivariate settings [1,2]. In parallel, the strong performance of simpler baselines has prompted the community to re-examine which inductive biases—rather than raw architectural complexity—truly matter for LSTF, especially when generalization across datasets is desired [1,9]. As a result, recent work has diversified along at least four axes: (i) rethinking attention and tokenization to better match multivariate structure and long contexts [1,2,6]; (ii) leveraging multi-resolution and frequency-domain structure to address non-stationarity and long-horizon dynamics [5,7,14,15]; (iii) learning scalable, time-varying dependency graphs for spatially or cross-channel correlated series [11,12,10]; and (iv) incorporating prior knowledge via foundation-model adaptation, language supervision, or synthetic-data priors to improve transfer and few-shot capability [3,4,17]. Alongside these routes, robust forecasting and uncertainty quantification have emerged as essential ingredients for long-sequence deployment, particularly under anomalies, missingness, and high-stakes decision requirements [9,10,18].

This survey adopts a narrative organization to emphasize methodological relationships rather than presenting isolated paper-by-paper summaries. Section 2 synthesizes the reviewed literature into coherent research trajectories and highlights how design choices propagate through efficiency, representation, and robustness considerations. Section 3 concludes with consolidated insights and forward-looking directions.

II. RELATED WORKS

A. Reconfiguring Attention and Sequence Representations for Long Context

iTransformer reframes a core assumption in Transformer-based forecasting: instead of treating each timestamp as a token whose embedding fuses multiple variates, iTransformer inverts the modeling axis and treats variates as tokens, allowing attention to focus on multivariate correlations while other components learn per-series representations [1]. This inversion directly targets two long-sequence pain points discussed in Transformer forecasters—performance degradation as lookback windows grow and the risk that mixed-variate temporal tokens yield uninterpretable or ineffective attention patterns when channels represent distinct

measurements or delayed effects [1]. From a narrative standpoint, iTransformer’s contribution is not an “efficient attention trick,” but a representational claim: long-context performance depends critically on aligning the tokenization granularity with the multivariate semantics.

VQ-TR addresses long sequences from a complementary angle by inserting vector quantization into the attention mechanism, effectively mapping long sequences to a discrete set of latent representations [2]. This design aims to reduce attention cost to linear complexity in sequence length while also providing a regularizing effect through discretization, which is particularly relevant when long-context models risk overfitting [2]. Notably, VQ-TR also foregrounds probabilistic forecasting and systematically compares Transformer-based methods under probabilistic evaluation, which reframes “long sequence” as not only a context-length issue but also a sample-efficiency and inference-time issue for uncertainty-aware prediction [2]. Together with iTransformer, VQ-TR illustrates a broader trend: recent LSTF research increasingly treats attention as a module whose internal representation (continuous vs. discrete, temporal vs. variate-centric) can be redesigned to reconcile long-context modeling with computational constraints.

SCAT extends the “rethinking Transformer design” thread by explicitly incorporating spectral structure and by altering how attention aligns with temporal progression. Its Spectral Central Alternating mechanism is motivated by the view that classical Transformer designs can ignore spatial information and noise considerations in multivariate series, and that attention matrices can be encouraged to reflect a sequence’s natural order rather than arbitrary cross-position interactions [6]. In relation to iTransformer and VQ-TR, SCAT can be read as targeting a different failure mode: even if attention is computationally feasible, long sequences can still be undermined by noisy correlations and misaligned dependency patterns, and architectural constraints can be used to bias the learned attention toward more plausible temporal dynamics [6].

B. Multi-Resolution and Frequency-Domain Modeling for Long-Horizon Dynamics

Multi-resolution modeling has emerged as a unifying response to two intertwined LSTF challenges: non-stationary dynamics and the coexistence of global trends with local fluctuations. DERITS operationalizes this idea in the frequency domain via Frequency Derivative Transformation (FDT), aiming to produce more stationary frequency representations and thereby mitigate distribution shift in non-stationary forecasting [5]. Its accompanying order-adaptive Fourier convolution and parallel multi-order design highlight a key intuition: long-horizon prediction benefits when the model can adaptively filter and combine frequency components at different derivative orders, rather than relying on a single-scale temporal representation [5]. In narrative terms, DERITS exemplifies a

“transform first, learn second” philosophy: reparameterize the series into a domain where long-range structure is easier to model, and then learn predictive mappings.

Wavelet-based Mixer architectures embody a closely related multi-resolution principle but instantiate it through wavelet decomposition and MLP-style mixing rather than Fourier operators. WPMixer combines wavelet decomposition, patching, and MLP mixing, arguing that multi-resolution wavelets can capture time–frequency information efficiently while patching extends the usable lookback window and stabilizes local information extraction [14]. WaveletMixer similarly leverages wavelets, but frames the approach as iterative, multi-level, multi-resolution, and multi-phase, and explicitly includes a Channel Learner to strengthen multivariate long-term forecasting [15]. The relationship between these models is informative: both use wavelets as a structured decomposition prior, yet they diverge in how they integrate channel interactions and training dynamics, suggesting that wavelet decompositions are becoming a flexible “front-end” upon which different mixing strategies can be layered [14,15]. TSKANMixer further supports the ongoing evolution of Mixer-style backbones by replacing standard MLP components with Kolmogorov–Arnold Network layers, implying that even within non-attention architectures, the functional form of mixing remains an active design dimension for improving accuracy across datasets [16].

mr-Diff brings multi-resolution structure into a generative modeling paradigm by adapting diffusion models to time series through seasonal–trend decomposition and a progressive, fine-to-coarse (and back) generation scheme [7]. Instead of treating time series diffusion as a straightforward port of image diffusion, mr-Diff uses decomposition to extract coarse trends first and then conditionally adds finer details, framing denoising as an “easy-to-hard” non-autoregressive process [7]. When viewed alongside wavelet-based mixers and DERITS, mr-Diff reinforces a broader methodological convergence: long-horizon forecasting is increasingly approached as a hierarchical reconstruction problem in which coarse components provide stable anchors and fine components are learned conditionally, improving both tractability and fidelity over long sequences.

C. Learning Graph Structures for Spatially Coupled and High-Dimensional Series

Graph-based approaches respond to LSTF settings in which multivariate series are not merely multiple channels of one system, but large collections of spatially or relationally coupled entities. TagSL (Time-aware Graph Structure Learning) focuses on the observation that correlations in cyber-physical systems can evolve over time with trends and periodicities, and that static graphs or unintentionally dynamic graphs may fail to represent such structured temporal variation [11]. By learning time-aware correlations from interactions of node and time representations, and by constraining learned correlations through time discrepancy learning and periodic discriminant functions,

TagSL positions graph learning itself as a long-sequence problem: the dependency structure must be temporally coherent across extended windows to support accurate forecasting [11].

SAGDFN pushes the graph-learning narrative toward scalability and adaptive diffusion in multivariate forecasting, emphasizing the need to precisely model complex correlations and shared patterns across many series while remaining computationally feasible at large node counts [12]. Its focus on graph diffusion as a forecasting mechanism places it in a complementary role to TagSL: where TagSL emphasizes temporally structured correlation learning, SAGDFN emphasizes scalable propagation and adaptive modeling over large relational systems, reflecting a practical constraint that becomes unavoidable in long-sequence, large-scale deployments [12].

BiTGraph highlights how real-world data quality issues reshape graph-temporal modeling. Rather than assuming intact observations or relying on imputation as a preprocessing step, BiTGraph integrates missingness into its modeling pipeline by injecting bias into temporal and graph modules to account for missing patterns, arguing that naive imputation can accumulate errors and degrade long-horizon performance [10]. In the context of TagSL and SAGDFN, BiTGraph illustrates an increasingly common theme: dependency learning is only as good as the observation process, and long sequences magnify the harm of systematic missingness unless the model explicitly internalizes missing patterns as part of the representation and propagation logic [10,11,12].

D. Foundation Models and Synthetic Priors for Transferable Forecasting

TEMPO frames time-series forecasting through the lens of generative pre-training, asking whether GPT-type architectures can serve as general-purpose time-series learners analogous to their role in language [3]. Rather than directly applying a language model to raw signals, TEMPO emphasizes two inductive biases: decomposition into trend/seasonal/residual components and prompt design for distribution adaptation across different series types [3]. This narrative is important for LSTF because long sequences often span regime shifts and heterogeneous domains; prompts become a mechanism for conditioning a general representation learner on dataset- or context-specific characteristics without retraining from scratch.

LeRet approaches the “language for forecasting” question from a parameter-efficient and modality-integration perspective. Instead of scaling up forecasting parameters to the size of an LLM, LeRet introduces a time-series retentive backbone augmented by a time-series-related language knowledge extractor and a modality integrator, aiming to distill pre-trained knowledge and causal-dependence modeling benefits into a forecasting model without large-scale parameter training [4]. Its emphasis on few-shot and zero-shot settings highlights a key emerging requirement in LSTF: accuracy is insufficient if adaptation

to new domains is expensive, and leveraging language descriptions can become a pragmatic route to encode semantics and priors when labeled histories are short [4].

TimePFN represents a different, but conceptually aligned, transfer strategy by learning a forecasting prior through synthetic data generation. By constructing diverse multivariate series via Gaussian process kernels and linear coregionalization, and training a transformer-based architecture to approximate Bayesian inference in a prior-data fitted network framework, TimePFN aims to achieve strong zero-shot and few-shot forecasting across benchmarks [17]. The relationship among TEMPO, LeRet, and TimePFN is revealing: all three treat cross-domain generalization as the central problem, but they operationalize “prior knowledge” differently—prompted foundation models (TEMPO), language-augmented retentive modeling (LeRet), and explicit synthetic prior learning (TimePFN) [3,4,17]. Collectively, they suggest that long-sequence forecasting is increasingly moving from dataset-specific fitting to meta-learning and prior-driven inference, motivated by the diversity of time-series applications and the scarcity of domain-specific training data.

E. Robustness, Missingness, Extremes, and Uncertainty Quantification in Long Sequences

Robustness concerns become unavoidable in long sequences because contamination effects propagate and amplify over time. RobustTSF formalizes time series forecasting with anomalies by statistically defining common anomaly types and analyzing loss and sample robustness under contamination, then proposing an efficient filtering-based algorithm inspired by learning with noisy labels [9]. Unlike pipelines that detect anomalies, impute, and retrain—approaches that can be brittle when detection is imperfect—RobustTSF argues for a more principled learning process that incorporates robustness into the training objective and sample selection logic itself [9]. In long-horizon settings, this orientation is particularly salient: robust learning is not a defensive feature but a prerequisite for stable multi-step extrapolation when training data is imperfect.

BiTGraph complements RobustTSF by focusing specifically on partial observability and missing patterns, integrating missingness into temporal and spatial modeling rather than treating it as a preprocessing artifact [10]. While RobustTSF targets anomalies as label/input contamination, BiTGraph targets structural missingness, and both reflect a shared methodological shift away from “clean then learn” workflows toward “learn under corruption” formulations that better reflect operational realities [9,10].

Extreme events introduce yet another failure mode for LSTF: rare but consequential values can dominate application risk, especially in hydrology and similar domains. The polar-representation-based extreme-adaptive model explicitly centers extreme values as first-class citizens in long-term forecasting, proposing representations and

learning strategies designed to capture both long-range dependencies and rare extremes [13]. This line of work underscores a practical insight for long sequences: as horizon increases, tail behavior often matters more than average behavior, and architectures and losses may need to be shaped accordingly.

Finally, long-sequence deployment often requires not only point accuracy but calibrated uncertainty. Neural Conformal Control proposes a neural conformal prediction method for time series designed to enhance adaptivity in non-stationary environments and to achieve target coverage, leveraging auxiliary multi-view data and monotonicity constraints for consistency across quantiles [18]. In narrative terms, conformal control reframes forecasting outputs from single trajectories to decision-ready intervals, and it situates calibration as an algorithmic design problem—particularly relevant when non-stationarity and long horizons make classical calibration assumptions fragile [18]. Taken together with robustness and missing-data modeling, uncertainty quantification completes an emerging picture: state-of-the-art LSTF is increasingly judged not only by average error but by its reliability under the messy conditions that long sequences inevitably encounter.

III. CONCLUSION

Long-sequence time-series forecasting has recently shifted from a narrow competition among backbone architectures into a broader design space where representation, decomposition, dependency structure, transfer learning, and reliability objectives interact. Across the reviewed works, architectural innovation often begins with a concrete diagnosis of why long contexts fail: attention may be misallocated when temporal tokens fuse heterogeneous variates, computation may become prohibitive as context grows, and spurious correlations can dominate when training data is limited or noisy. Reconfiguration-based models address these issues by changing the unit of modeling (e.g., variate-centric tokenization) or by compressing attention through discretization, aiming to preserve long-context capacity while reducing overfitting and complexity.

A second recurring innovation is the re-emergence of multi-resolution structure as a primary inductive bias. Wavelet and frequency-based methods explicitly decompose signals so that global trends and local fluctuations are modeled at compatible scales, while diffusion-based forecasting uses progressive generation from coarse to fine to stabilize long-horizon synthesis. These strategies collectively suggest that long-horizon extrapolation benefits when learning is organized hierarchically, with coarse components acting as anchors that reduce downstream uncertainty in fine-scale reconstruction.

Multivariate dependency learning has likewise matured beyond static correlation graphs. Time-aware structure learning and scalable diffusion mechanisms treat the graph as a dynamic, learnable object whose evolution must be coherent over time, and missing-value-aware designs integrate the observation process into temporal and spatial

modeling rather than relying on imputation. Meanwhile, transfer-oriented paradigms—prompted foundation models, language-augmented forecasting, and synthetic-prior training—indicate a growing emphasis on generalization across domains and few-shot adaptation, reflecting real-world constraints where long labeled histories are not always available.

Looking forward, several directions appear especially promising: unifying multi-resolution decomposition with adaptive dependency graphs; developing training objectives that jointly optimize long-horizon accuracy, robustness to corruption, and calibrated uncertainty; and scaling transferable forecasters that can leverage external knowledge without sacrificing efficiency. Ultimately, the most deployable long-sequence forecasters will likely be those that integrate structured inductive biases with principled reliability guarantees, rather than those that optimize point accuracy alone.

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